

AI Ops/Governance

#	Bài toán cụ thể	Function / journey	Data signal	AI/DS/ML	Output	Được gì	Rủi ro / mất gì
1	Prediction logging for fintech model	Model inference	input features, prediction, confidence, model version, timestamp	DS/MLOps-lite	prediction log	Nền tảng cho monitoring/governance	Không nên làm standalone
2	Model version audit trail	Model deployment/inference	model version, training data version, deployed_at, request_id	SE + DS audit	model lineage record	Học auditability, rất fintech	AI/ML nhẹ nếu đứng riêng
3	Feature drift detection	Model monitoring	feature distribution train vs current	DS drift metrics	drift alert	Rất thực tế cho AI governance	Cần data theo thời gian
4	Prediction drift detection	Model monitoring	prediction score distribution over time	DS/statistics	prediction drift alert	Dễ demo hơn label-based monitoring	Drift không đồng nghĩa model sai
5	Label delay monitoring	Sau khi actual outcome đến muộn	prediction time, label available time, missing label rate	DS analytics	label delay report	Rất thực tế với credit/fraud	Cần mô phỏng delayed labels

6	Model performance monitoring	Sau khi có labels	prediction, actual outcome, time window	DS metrics	AUC/F1/precision/recall trend	Rõ DS/ML, mentor dễ góp ý	Cần labels synthetic hoặc delayed
7	Threshold performance simulator	Decision policy tuning	prediction scores, actual labels, threshold	DS evaluation	precision/recall trade-off chart	Học decision threshold, rất quan trọng	Không phải product nếu đứng riêng
8	Calibration monitoring	Probability model review	predicted probability, actual outcome buckets	DS calibration	calibration curve/report	Rất hay cho credit/risk model	Hơi nâng cao
9	False positive workload monitor	Human review queue	model decision, actual label, reviewer workload	DS analytics	false positive/review load report	Gắn model với ops thực tế	Cần giả lập reviewer queue
10	False negative risk monitor	Risk model review	missed fraud/default cases, model score	DS analytics	missed-risk report	Học trade-off risk	Need labels, có thể nhạy cảm
11	Human override analysis	Reviewer khác model	model score, reviewer decision, actual outcome	DS analytics	override pattern report	Rất hay cho human-in-the-loop	Cần mô phỏng reviewer decisions

12	Reviewer-model disagreement detector	Decision support workflow	model decision, reviewer decision, risk signal	DS/anomaly	disagreement alert	Hợp credit/KYC/fraud review	Không phải model core
13	Decision audit log checker	Compliance logging	user_id, model_version, decision, reason, timestamp	Rule + DS	missing audit field alert	Thực tế trong fintech governance	AI/ML nhẹ
14	Explainability coverage monitor	Model explanation quality	prediction, reason codes, SHAP values, missing explanations	DS/rule	explanation coverage report	Đảm bảo quyết định có reason	Không đánh giá explanation đúng/sai sâu
15	Explanation stability test	XAI validation	similar inputs, explanations, feature importance variance	DS/XAI evaluation	unstable explanation alert	Rất hay, tránh explainability “cho có”	Khó hơn, cần hiểu XAI
16	Reason code generator for credit decision	Credit decision explanation	model factors, policy rules, applicant data	XAI + LLM structured output	reason codes	Hợp credit governance	Cần tránh legal overclaim
17	LLM explanation groundedness checker	AI-generated explanation review	explanation text, source facts, model features	LLM eval + rule checks	grounded / unsupported claims	Rất hợp AI governance	LLM eval có thể không đáng tin tuyệt đối

18	Hallucination detector for ops assistant	AI assistant response	generated answer, source docs, case facts	RAG eval/LLM judge + rules	hallucination risk	Hợp RAG/AI ops assistant	Evaluation khó, cần manual sample
19	RAG citation coverage monitor	RAG assistant	answer, retrieved docs, citations, query	RAG eval	citation coverage score	Học grounded AI	Dễ trở thành RAG project phụ
20	Retrieval quality evaluator	RAG pipeline	query, retrieved chunks, expected docs	IR metrics	precision@k / recall@k	Rõ methodology	Cần labeled query-doc pairs
21	Prompt regression testing	LLM workflow update	test cases, prompt version, output diff	AI eval	pass/fail prompt test	Rất thực tế AI app engineering	Không fintech-specific nếu thiếu workflow
22	Guardrail violation monitor	LLM assistant runtime	prompt, output, policy rules, blocked content	Rule/LLM eval	violation alert	Học safety/guardrails	Cần define policy
23	PII leakage detector in AI output	AI response validation	generated text, PII patterns, source data	NLP/rule	PII leakage flag	Cực quan trọng trong bank/ví	Có thể rule-heavy

24	Sensitive data minimization checker	Model/AI input logging	input payload, PII fields, logs	Rule + DS audit	unnecessary data warning	Học privacy-by-design	Không phải ML core
25	Customer notification compliance checker	AI interaction	channel, customer-facing AI flag, consent/notice	Rule + audit	missing AI notice alert	Bám regulatory mindset; SBV draft yêu cầu notify khi AI trực tiếp tương tác khách hàng	Cần nói là theo public draft/signal, không production compliance
26	AI interaction disclosure logger	Chatbot/customer AI	user session, AI disclosure shown, timestamp	SE audit + DS	disclosure audit log	Rất thực tế với AI chatbot	AI/ML nhẹ
27	Bias/fairness dashboard for credit model	Model governance	approval rate, error rate by segment	DS fairness metrics	fairness report	Rất hợp responsible AI	Nhạy cảm, phải dùng synthetic groups
28	Fairness drift monitoring	Model over time	group-level prediction/error trend	DS monitoring	fairness drift alert	Có chiều sâu hơn fairness static	Cần data theo thời gian
29	Segment performance monitor	Risk model ops	metric by region/channel/device/customer segment	DS analytics	weak segment report	Thực tế, ít nhạy cảm hơn fairness	Cần segment labels

30	Out-of-distribution input detector	Model inference	feature distance, missing rate, unseen category	Anomaly/OOD detection	OOD warning	Rất hữu ích cho model safety	Hỏi nâng cao
31	Low-confidence decision routing	Human-in-the-loop	prediction probability, entropy, missing data	ML uncertainty/rule	route to human review	Gắn model với ops, rất thực tế	Cần define confidence threshold
32	Auto-approve safety gate	High-stakes decision	risk score, confidence, policy rules	Rule + ML guardrail	auto / review / block	Giúp tránh model tự quyết quá mức	Cần careful framing
33	Model input data quality monitor	Data pipeline → model	missing values, invalid ranges, stale features	DS/rule	data quality alert	Rất thực tế MLOps-lite	Không quá “AI” nếu đứng riêng
34	Feature freshness monitor	Real-time/batch features	last updated time, stale feature count	DS/rule	stale feature alert	Học production feature pipeline	Cần simulate feature store
35	Training-serving skew detector	Model serving	train feature stats vs serving feature stats	DS comparison	skew alert	MLOps concept quan trọng	Hỏi technical, ít domain nếu đứng riêng

36	Model rollback decision helper	Model incident	performance drop, drift, incident severity	Rule + DS	rollback recommendation	Học AI operations	Cần deployment simulation
37	Champion-challenger model comparison	New model evaluation	current vs candidate metrics, drift, fairness	DS/ML evaluation	promote / reject recommendation	Rất đúng model lifecycle	Có thể hơi nhiều cho 5 tuần
38	A/B test monitor for model decision	Experiment	treatment/control outcome, conversion/default/fraud rate	DS experiment analysis	experiment result	Học product/data experimentation	Cần simulated experiment data
39	Model card generator	Model documentation	dataset, metrics, limitations, intended use	LLM structured generation + template	model card draft	Học documentation/governance	Không nên standalone
40	Dataset card / data lineage generator	Data governance	dataset source, schema, preprocessing, known bias	LLM/template + DS metadata	dataset card	Có governance maturity	Ít product nếu đứng riêng
41	Risk assessment checklist for AI use case	Before deploying AI	use case type, user impact, data sensitivity	Rule + LLM checklist	risk assessment report	Rất hợp mentor discussion	Có thể quá process-heavy

42	AI use-case inventory dashboard	Organization AI governance	model/use case, owner, risk tier, data type	SE + DS analytics	AI inventory	Enterprise governance mindset	Không phù hợp 5-week product nếu standalone
43	High-risk AI use case classifier	AI governance intake	decision impact, automation level, data sensitivity	Rule/ML-lite	risk tier	Học risk classification	Phụ thuộc policy giả lập
44	Policy-to-control mapping assistant	Governance ops	policy docs, system facts, controls	RAG/LLM structured mapping	required controls list	Hợp AI governance assistant	Dễ abstract
45	Evidence collection tracker	AI audit preparation	test results, logs, model card, approvals	SE workflow	evidence completeness	Học audit readiness	AI/ML nhẹ
46	Responsible AI checklist validator	Before release	model metrics, fairness test, explainability, monitoring	Rule + LLM	release readiness score	Rất practical add-on	Không nên làm chính
47	AI incident detector	Runtime monitoring	spikes in errors, complaints, model drift, hallucination flags	DS/anomaly	AI incident alert	Học AI ops nghiêm túc	Cần multi-signal data

48	AI incident postmortem generator	Sau AI incident	timeline, metrics, root cause, actions	LLM structured report	postmortem draft	Hợp ops/engineering culture	Cần evidence, không hallucinate
49	LLM cost and latency monitor	AI app runtime	token usage, latency, model, endpoint	DS analytics	cost/latency dashboard	Rất thực tế AI app engineering	Không fintech-specific
50	LLM fallback policy evaluator	LLM production workflow	model errors, latency, confidence, fallback result	Rule + DS	fallback decision quality	Học AI reliability	Cần simulate model calls
51	Tool-call trace analyzer for AI agent	Agent workflow	tool calls, errors, retries, latency, result validity	DS/log analysis	problematic trace report	Hợp agent/ops bạn đã học	Agentic scope dễ phình
52	Agent permission / least-privilege checker	AI agent security	tools accessible, actions executed, user role	Rule + security audit	excessive permission alert	Rất mới và thực tế; finance chú ý agent security	Có thể xa ML/DS
53	Agent action audit trail	AI agent interacting with systems	action, tool, user, timestamp, result	SE audit + DS	traceable action log	Hợp secure AI ops	Không nên standalone

54	Prompt injection attempt detector	RAG/agent input	user prompt, retrieved text, suspicious patterns	NLP/rule/ML-lite	injection risk flag	Rất hợp AI assistant safety	Cần synthetic attacks
55	RAG source poisoning detector	Knowledge base ingestion	document source, change history, abnormal content	DS/anomaly/rule	suspicious source alert	Có security angle sâu	Khó nếu thiếu KB workflow
56	AI-generated financial advice guardrail	Customer-facing AI	answer text, prohibited advice patterns, risk terms	Rule/LLM classifier	block/rewrite/review	Rất fintech-relevant	Cần define policy carefully
57	AI output tone/compliance checker	Customer support AI	response text, policy, tone rubric	LLM eval + rules	compliant/non-compliant	Practical for chatbot	Evaluation subjective
58	Customer complaint signal monitor for AI chatbot	Post-AI interaction	complaint text, rating, escalation, topic	NLP/DS analytics	complaint trend	Học feedback loop	Cần simulated/support data
59	Model feedback loop quality checker	Human feedback collection	reviewer labels, disagreement, correction reason	DS analytics	feedback quality score	Rất hay cho human-in-loop	Cần review data

60	Governed AI workflow dashboard	End-to-end AI ops	logs, metrics, drift, audit, guardrails, incidents	DS + SE dashboard	governance dashboard	Có thể gộp nhiều module	Quá rộng nếu làm project chính
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Package 1 — Model Monitoring & Decision Governance

Gồm:

#1 Prediction logging
 #3 Feature drift detection
 #4 Prediction drift detection
 #6 Model performance monitoring
 #7 Threshold performance simulator
 #31 Low-confidence decision routing

Package 2 — Responsible Credit AI Governance

Gồm:

#16 Reason code generator
 #27 Bias/fairness dashboard
 #28 Fairness drift monitoring
 #29 Segment performance monitor
 #34 Human override analysis
 #35 Credit decision audit trail

Package 3 — RAG/LLM Safety & Evaluation for Fintech Ops Assistant

Gồm:

- #17 LLM groundedness checker
- #18 Hallucination detector
- #19 RAG citation coverage monitor
- #20 Retrieval quality evaluator
- #22 Guardrail violation monitor
- #23 PII leakage detector
- #54 Prompt injection detector

Package 4 — AI Agent Governance & Traceability

Gồm:

- #51 Tool-call trace analyzer
- #52 Agent permission checker
- #53 Agent action audit trail
- #54 Prompt injection attempt detector
- #49 LLM cost/latency monitor

Package 5 — AI Release Readiness / Governance Checklist

Gồm:

- #39 Model card generator
- #40 Dataset card generator
- #41 Risk assessment checklist
- #43 High-risk AI use case classifier
- #46 Responsible AI checklist validator
- #45 Evidence collection tracker

